

CRM ANALYSIS FOR MAVEN ANALYTICS

1. Overview

Maven is a B2B computer hardware business that has recently implemented a CRM system to manage their sales activities. However, the company has not realized the full potential of CRM in monitoring performance of their sales team.

The purpose of this project is to develop an interactive dashboard that

- (1) informs executive team of the company's current sales and performance
- (2) enables sales managers to understand how their team contribute to the company KPI and track the team's progress.

2. Dataset

The dataset contains records exported from MavenTech's CRM from October 2016 to December 2017. It has details of opportunities with associated information such as product, account, and the stage the sales in (prospecting, won, lost, engaging).

3. Thought process

Before creating a dashboard, I always starts with gathering requirements to understand the key metrics that stakeholders are interested in and business questions they want to answer. Since this step is unable to obtain with this project, I derive the key questions based on my previous experience interacting with C-level executive team and sales managers.

From the perspective of an executive level member, they would like to understand if the company's sales revenue is healthy. Additionally, they also want to gain key insight into areas the company should invest in to build a strong pipeline.

- 1. Which team is performing the best/ worst? Do they need support in terms of workload?
- 2. Where is the sales hot spot that the company should invest more resources in?
- 3. What industry/ clients that buy the most from us so that we can organize more relevant events to cultivate stronger relationship with them?

For a sales manager, they usually would like to know how their team contributes to the company's main goal. The questions to answer are:

- 1. How is my team tracking against the company KPIs?
- 2. How is my team performing compared to the rest of the organization?

3. How are my sales agents performing compared to each other? Are there team members who require extra support in certain areas?

4. Where should I focus my efforts for the rest of the quarter to improve performance? (which deals, products, etc.)

4. Metrics

In terms of metrics, the most important company KPI is revenue generation. Related metrics such as number of sales, average sales value, conversion time and price discount are also necessary to understand the overall health of the sales pipeline. I captured all these metrics in one single page so executive team and sales managers have a complete visibility and context when reviewing the company/team performance.

i/ Total Sales/ Wins

These metrics are fairly self explanatory. There are sales leads that are either still in some sort of engagement, or have been won or lost. Here, I just wanted to track the wins. This is a very basic metric, but can give a basic idea of magnitude of performance, especially when looking at QoQ.

ii/ Average Days to close

This metric measures the average number of days between engagement and closing a winning deal, essentially an "efficiency" metric. The shorter the timeframe required to close a deal, the more time the agent has to pursue other deals.

iii/ Average Deal Value

This looks at the average dollar value of a successful sale. Again, this is important in order to provide full context. A sales agent may have 40 wins, but only takes in a \$50 average for each deal. Another agent might have 5 wins, but averages \$3000 per sale.... who is the better performer? This metric can provide that counterbalance and context.

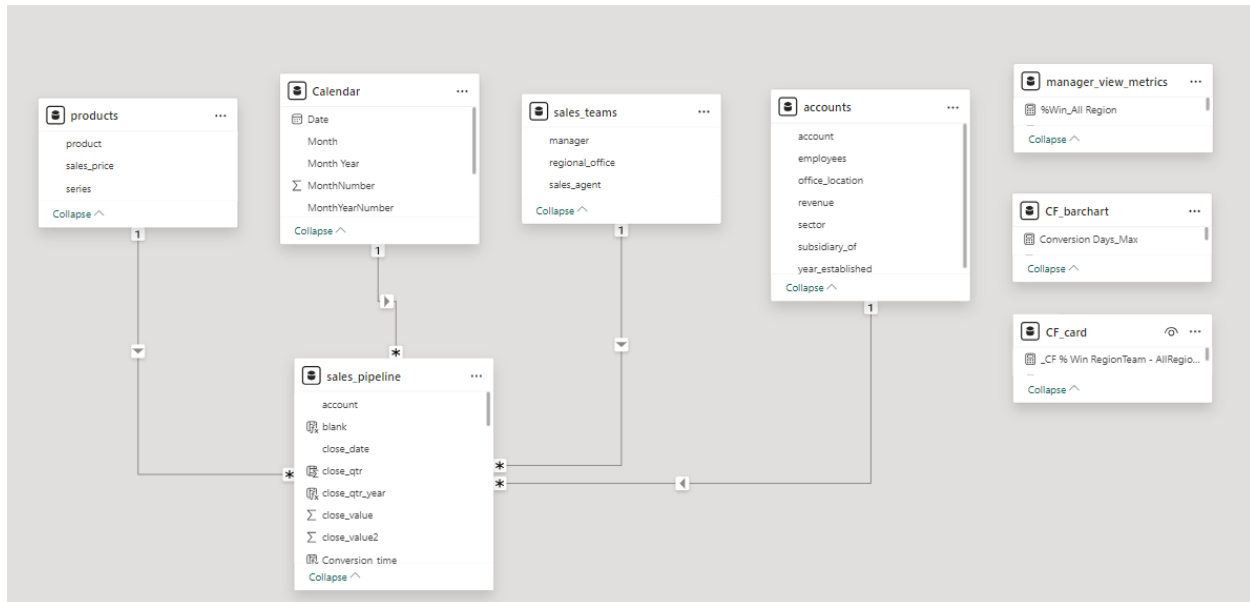
iv/ Average Discount

Finally, in order to assess if the agents are potentially underselling goods in order to make a deal, I provided another metric named Average Discount Price to measure the variance of the actual sales price from the Recommending Selling Price. Usually, there might be a target limit the manager sets for the team, especially if the product is positioned as premium in the market.

One noteworthy point here is win percentage is the least important metric for now. The reason is sales team generally only enter opportunities they are more likely to win, which artificially boost their hit rate. I think this concept applies to Maven too since based on the current hit rate of 60.3%, the company is holding more than half of the market. This might be too high, given the competitiveness of the software industry and the relatively small size of the company. However, in reality, I will discuss more with the sales managers if this assessment aligns with their market understanding.

5. Data Model

The data model is straightforward where sales_pipeline is the source table that connects to products, sales_teams and accounts tables. Calendar table is created for the ease of quarter calculations. Additional metrics table are manager_view_metrics, CF_barchart and CF_card are created to provide context to different sales related metrics.



6. The dashboard

The most crucial criteria in designing a dashboard is to make sure the users understand the purpose and how to use the dashboard. Therefore, I provided some notes in each page to explain how key metrics are calculated. This page helps the users navigate the dashboard and interpret the information confidently.

The dashboard sections are explained as below.

Executive Summary page

This page provides high level insights on the company current performance.

Firstly, it is important to define what the metrics mean on the report. This was done via smaller font as a note. Potentially, as the users get more familiar with the report, these could be removed, or moved to a notes page.

MAVEN ANALYTICS
QUARTERLY PERFORMANCE DASHBOARD

QTR4

Wins refer to sales contracts closed within the reporting quarter.

Win % refers to the number of contracts won as a percentage of wins and losses.

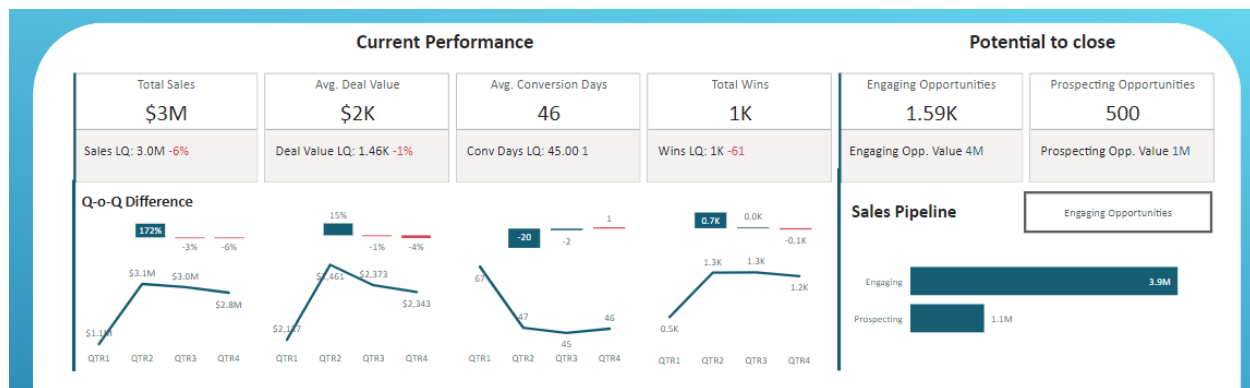
Average Days to Close is the average days between engagement and contract close for projects that have been won.

Average Sales Value is the average value of contracts won.

Average Price Discount refer to the average price difference between sales price and recommended wholesale price (RWP)

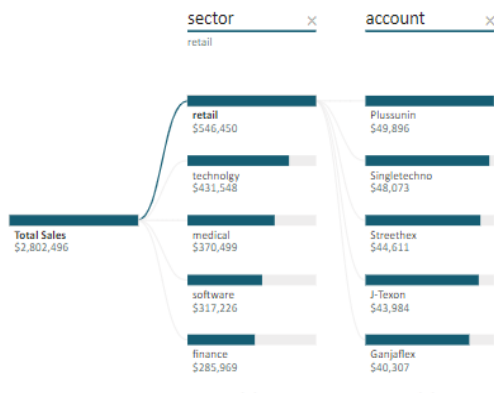
The key KPIs are shown clearly at the top of the page, with figures for the previous quarter and Quarter over Quarter performance (QoQ) for context.

They are grouped into the current performance and potential to close. The current performance includes metrics such as Total Sales, Total Wins, Average Deal Value and Average Conversion Days. Potential to close contains opportunities that sales teams are working on and likely to close in the next 1-2 quarters.



Below the page, users can click have a deep dive view into the sales breakdown for the current quarter. This helps them understand biggest industries, accounts, and products the company are selling to, hence organizing the right marketing campaigns to attract these groups. Furthermore, from the performance of sales teams, leaders can also have a feel of stronghold regions that the company should increase their focus on.

Biggest Sectors and Accounts



Best Selling Products

product	Total Sales	#Won	Avg. Deal Value	Conv. Days
GTK 500	\$83,377	3	\$27,791	24
GTX Basic	\$139,659	27	\$513	46
GTX Plus Basic	\$193,447	178	\$1,087	44
GTX Plus Pro	\$795,530	143	\$5,563	44
GTXPro	\$958,481	200	\$4,792	45
MG Advanced	\$619,162	183	\$3,383	47
MG Special	\$12,830	232	\$55	47
Total	\$2,802,496	1196	\$2,343	46

Best Performing Teams

manager	Total Sales	#Won	Avg. Deal Value	Conv. Days
Melvin Marxen	\$670,788	243	\$2,760	44
Summer Sewald	\$559,649	248	\$2,248	46
Rocco Neubert	\$513,705	175	\$2,935	48
Celia Rouche	\$430,212	179	\$2,403	45
Dustin Brinkmann	\$316,994	213	\$1,488	49
Cara Losch	\$311,148	137	\$2,271	47
Total	\$2,802,496	1196	\$2,343	46

Team Performance page

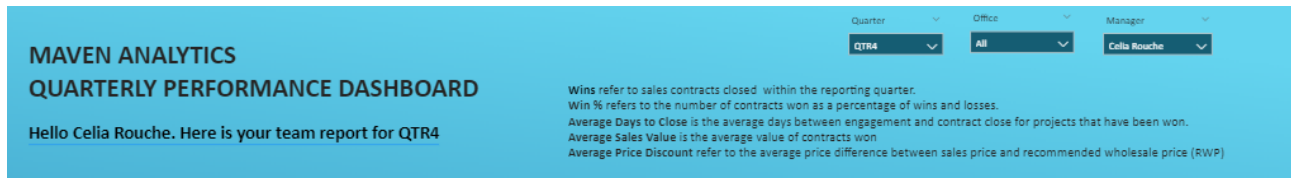
There are 3 sales managers for each region, so total 6. Each of them has 5-6 agents reporting to them. In order to understand how the team perform, we need to establish some benchmark for the sales manager to compare themselves against. Therefore, for each of the sales metrics, I show it on four different levels:

- global level
- regional level
- team level
- agent level

For these visuals, a manager can select themselves and any given quarter period and see the results for that quarter, as well as Q-o-Q performance.

Page Design

The header is a place where a manager select their team, quarter period they want to view. I also wanted to provide some kind of dynamic personalized greeting for the manager, and confirm the team details. That way if they needed to screen shot the report, the details would be there. The dynamic parts of the greeting were done using SELECTEDVALUE() type measures.



Card Visuals

The metrics were displayed in the new card visuals, and formed a consistent hierarchy of:

- All Regions
- Team Region
- Team

	Total Sales	Wins	Win %	Average Deal Value
All Regions	3M	1.2K	60.3%	2.3K
Team Region	990K	428	63.4%	2.3K
	35% of All Regions	36% of All Regions	▲ +3.2% vs. All Regions	▼ -1.3% vs. All Regions
Selected Team	430K	179	61.7%	\$2K
	15% of All Regions	15% of All Regions	▲ +1.5% vs. All Regions	▲ +2.6% vs. All Regions
	43% of Team Region	42% of Team Region	▼ -1.7% vs. Team Region	▲ +3.9% vs. Team Region
	▼ QoQ difference:-12.3%	▲ QoQ difference:+5 wins --	▲ QoQ difference:+1.5%	▼ QoQ difference:-\$416

To achieve these type of metrics, I employed the below DAX, using Average Deal Value as an example. I use "REMOVEFILTERS()" to exclude the selected filters for All Regions and Team Region. For example, in the case of Average Deal Value,

Average Deal Value for All Regions

```
Deal Value_All Region =  
CALCULATE(  
[deal_value],  
REMOVEFILTERS(sales_teams[regional_office], sales_teams[manager],  
sales_teams[sales_agent])  
)
```

```

Deals Value_Region =
var _Region = SELECTEDVALUE(sales_teams[regional_office])
var _Value =
CALCULATE(
    [deal_value],
    sales_teams[regional_office] = _Region,
    REMOVEFILTERS(sales_teams[manager], sales_teams[sales_agent])
)

```

To add the dynamic context references, I employed a few measures that using variables and SWITCH(TRUE()). For example,

Average Deal Value by the Team vs All Regions

```

Deal Value RegionTeam- All =
VAR _Value =
    [Deals Value_Region] / [Deal Value_All Region] -1
RETURN
SWITCH(TRUE(),
    _Value > 0 , "▲ " & FORMAT ( _Value, "+0.0%;-0.0%") & " vs. All Regions" ,
    _Value < 0 , "▼ " & FORMAT ( _Value, "+0.0%;-0.0%") & " vs. All Regions" ,
    _Value = 0, "No difference"
)
RETURN
    if(_Value = BLANK(), "-", _Value)

```

Average Deal Value by the Team vs Team Region

```

Deal Value Team- TeamRegion =
VAR _Value =
    [deal_value] / [Deals Value_Region] -1
RETURN
SWITCH(TRUE(),
    _Value > 0 , "▲ " & FORMAT ( _Value, "+0.0%;-0.0%") & " vs. Team Region" ,
    _Value < 0 , "▼ " & FORMAT ( _Value, "+0.0%;-0.0%") & " vs. Team Region" ,
    _Value = 0, "No difference"
)
RETURN
    if(_Value = BLANK(), "-", _Value)

```

Colour formatting was applied as an additional step based on whether the result was positive or negative, for example

```

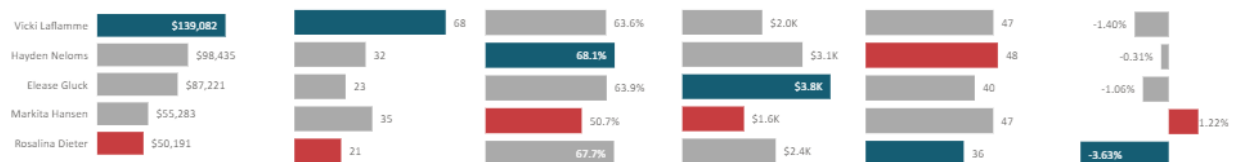
_CF Deal Value Team - Region =
VAR _Value = [deal_value]/ [Deals Value_Region] -1
Return
SWITCH(TRUE(),
    _Value > 0 , "#82b269",
    _Value < 0 , "#c73d40",
    "#aaaaaa")

```

Bar Chart

The bar chart displays selected team's sales agents by different metrics. The order of the sales agents is controlled so managers dont have to jump up and down to look for the agents they want.

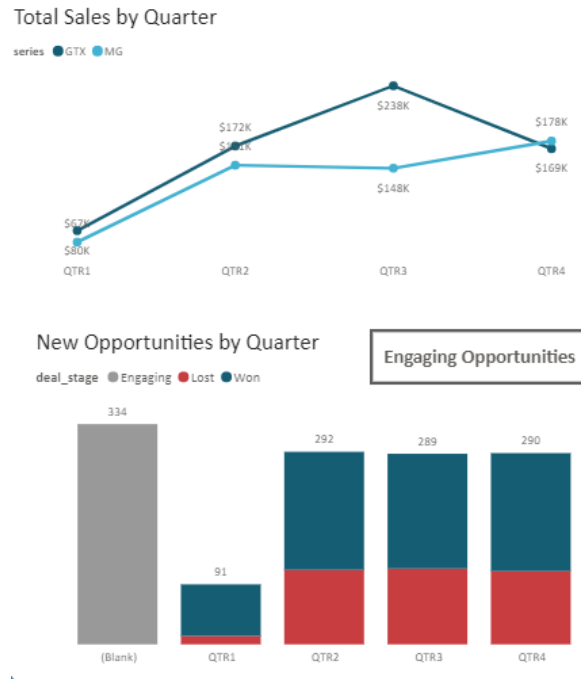
Agent Details



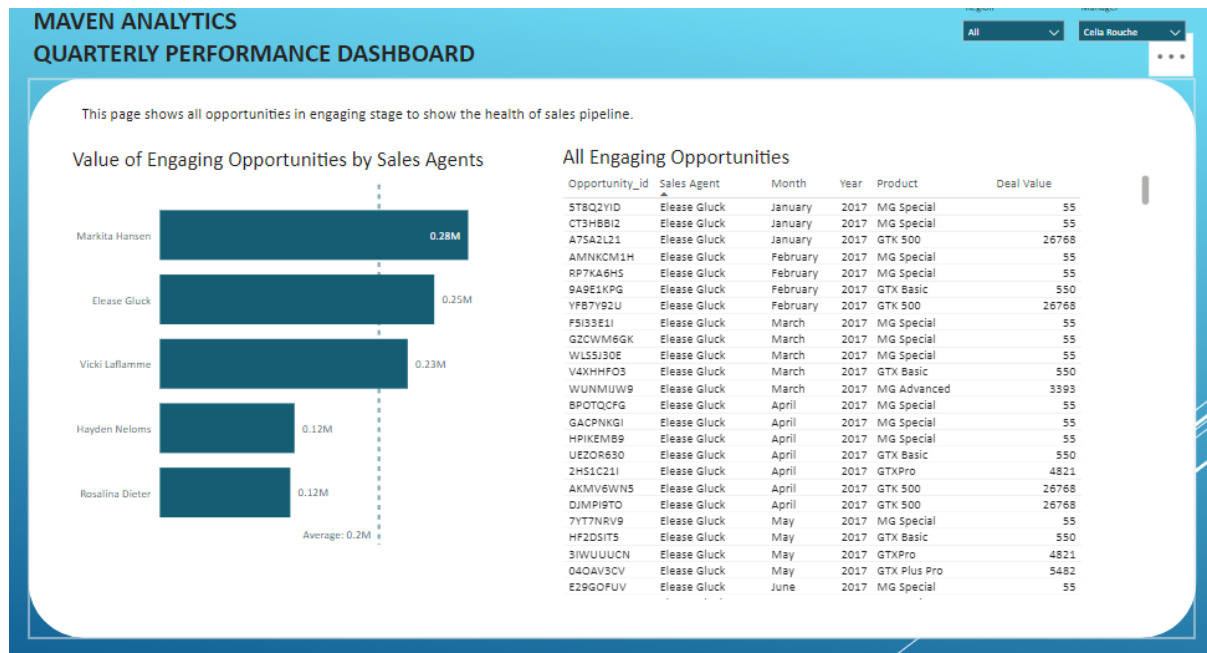
Product Analysis page

From the sales overview captured by the Team Performance tab, managers would then like to see which products their teams can sell well so they can support their team accordingly. Therefore, I designed a product analysis page that shows product series and products sold by each team.

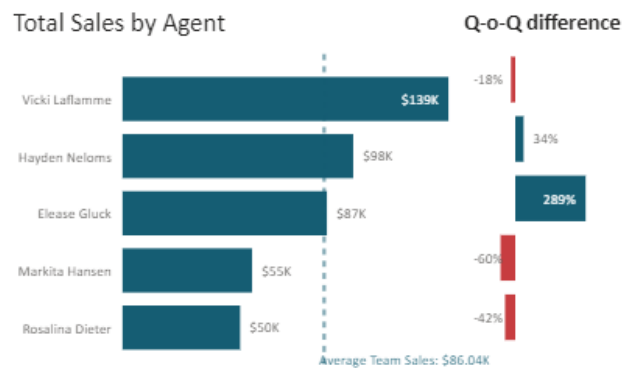
Charts on the left show product performance as a whole.



Like the Executive Summary page, I provide a button that leads to the Engaging Opportunities here. Sales managers can use this detail to prioritize actions such as following up offers to close before end of quarter, and CRM cleanups, eg opportunities that have been open for too long and need to be marked as lost.



Charts on the right display product performance by sales agents. From here, we can see the strengths/weaknesses of each agent based on the products they sold.



Agent Product Performance for selected sales metric and selected quarter

sales_agent	GTX Basic	GTX Plus Basic	GTX Plus Pro	GTXPro	MG Advanced	MG Special
Vicki Laflamme	\$10,040	\$2,099	\$32,034	\$43,375	\$50,592	\$942
Hayden Neloms		\$5,485	\$21,806		\$71,035	\$109
Markita Hansen	\$3,732	\$7,790	\$0	\$24,023	\$19,138	\$600
Elease Gluck	\$1,048	\$953	\$0	\$10,293	\$17,150	\$586
Rosalina Dieter	\$1,009	\$920		\$4,614	\$16,876	\$586
Total	\$15,829	\$17,247	\$53,840	\$82,305	\$174,791	\$2,823

If time permits, I can improve this page by showing other sales related metrics such as Wins, Average deal size, Average Price Discount. This can be done using field parameters to allow selection of preferred metric.

7. Conclusion

CRM is a valuable source of data for the executive level and sales teams to understand the company's sales performance, their strengths and areas for improvement in terms of products and accounts. This dashboard provides a comprehensive view of the current standing as well as potential opportunities to leverage on.